The High-Redshift Accretion Atlas

A Tour of the Final Catalogue and Paper-Ready Results

Repository status document

2 September 2026

Abstract

This document is a guided tour of the repository's most significant scientific products and the pipeline that produces them. Versions now identify nested datasets rather than software releases. v1 is the complete analysis of the original 23-object JADES broad-line AGN catalogue; v2 applies the same analysis to 112 comparable broad-line AGN; and v3 is the final heterogeneous atlas of 234 measurements, 219 physical objects, and 218 host systems. Every version has the same baseline evaluation, ranking, 10,000-sample Monte Carlo uncertainty, duty-cycle, compatibility, and visual products. In v3, 196 objects support numerical growth inference and 23 remain visible as explicit no-inference cases. The figures are diagnostic comparisons under stated assumptions, not evidence for a unique seed or accretion history.

1 Start here: the three dataset views

The central design rule is simple: changing version changes catalogue membership, not the scientific question or the visual language.

Table 1: Canonical dataset scopes and analysis coverage.

Version	Scope	Measurements	Objects	Hosts	Numeric objects
v1	Original JADES BLAGN	23	23	23	23
v2	Expanded comparable BLAGN	119	112	111	112
v3	Final heterogeneous atlas	234	219	218	196

The canonical one-row-per-object tables live in `data/processed/v1/`, `data/processed/v2/`, and `data/processed/v3/`; each is named `<version>_accreting_objects.csv`. The corresponding measurement tables retain every literature measurement, including alternates of the same physical object. Identity products in `data/crossmatch/<version>/` record aliases, preferred measurements, object-host links, and reviewed match decisions.

v2 groups the sources that use a comparable broad-line workflow: JADES, Taylor CEERS/RUBIES, Matthee EIGER/FRESCO, Lin ASPIRE, Harikane NIRSpec, Davis/THRILS, and Ren ALPINE/CRISTAL. v3 adds the object types that require materially different handling: XQR-30 and GNIRS-50 luminous quasars, the UHZ1 X-ray evidence history, and the audited Scholtz narrow-line candidates.

2 What is in the final catalogue?

Figure 1 is the quickest scientific overview. The left panel shows every growth-eligible preferred object in redshift-mass space. The right panel makes the analysis boundary explicit: objects without a supported canonical mass stay in the catalogue instead of disappearing from the figure.

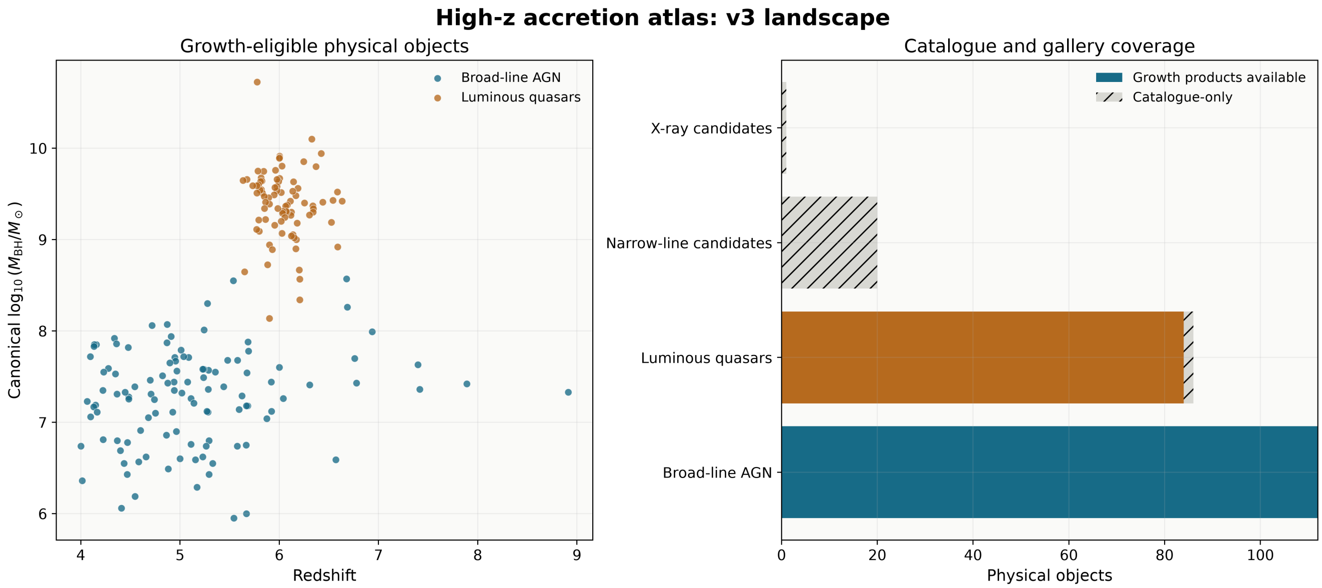

Figure 1: Final v3 catalogue landscape. The heterogeneous classes are kept visually distinct, and catalogue-only objects are counted rather than assigned an inferred mass.

The final object composition is 112 broad-line AGN, 86 luminous-quasar comparison objects, 20 narrow-line candidates, and one X-ray candidate. The catalogue contains 219 unique physical objects even though it retains 234 measurements. This difference is scientifically useful: alternate measurements are preserved for sensitivity tests, while one documented preferred row drives each object-level result.

How it was obtained. Source-specific ingestion code preserves native measurements and provenance. The shared catalogue layer then normalizes vocabulary, resolves physical identities and hosts, applies the preferred-row policy, and computes evidence, mass-comparability, lensing, conditional-mass, and growth-eligibility fields. The public `scripts/00_process_catalogues.ipynb` notebook calls the tested `src/internal/process_catalogues.py` implementation to materialize v1, v2, and v3 from this one corrected pipeline.

3 The physical calculation behind every result

The atlas uses the Dayal-style exponential growth relation

$$M_{\rm BH}(t_{\rm obs}) = M_{\rm seed} \exp\left[f_{\rm Edd} \left(\frac{1-\epsilon}{\epsilon} \right) \frac{\Delta t}{t_{\rm Edd}} \right] B_{\rm merge}, \tag{1}$$

with cosmic time from a flat Planck-style cosmology. The model is evaluated in three directions: predict the final mass, solve for the required lifetime-average $f_{\rm Edd}$ at fixed seed mass, and solve for the required seed mass at fixed accretion history. The baseline ranking uses $z_{\rm seed} = 30$, $\epsilon = 0.1$, and $B_{\rm merge} = 1$.

Figure 2 places the final masses against representative reference tracks. It is a timing and scale diagnostic, not a fit. The lower strip shows all 23 objects for which a numerical growth track is deliberately unavailable.

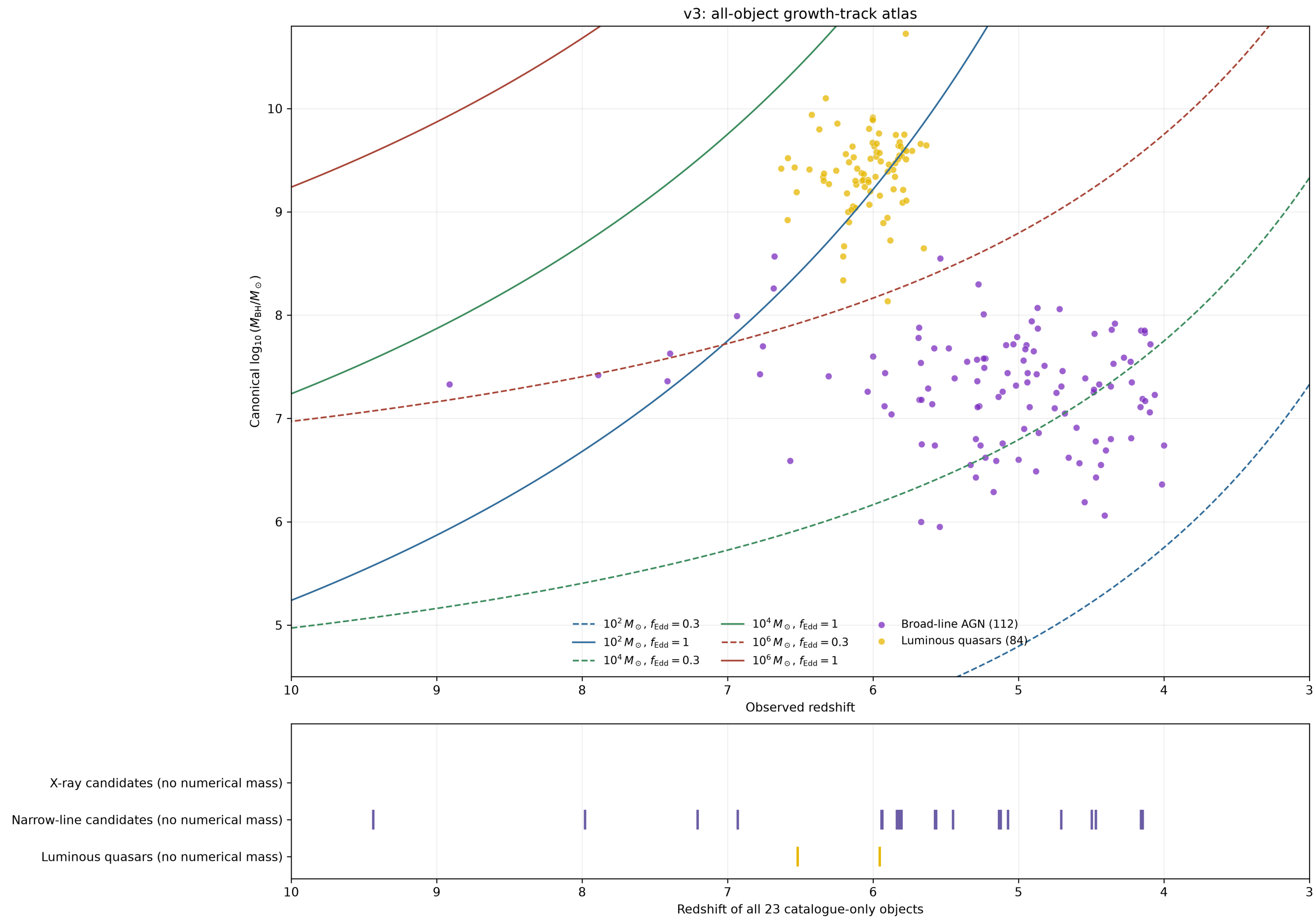

Figure 2: All-object growth-track view. Supported masses appear above; every unsupported object remains visible below with its class and redshift. The common reversed redshift axis spans $z = 10$ to $z = 3$ with unchanged figure dimensions and margins. Broad-line AGN are purple and luminous quasars are yellow.

A separate comprehensive companion preserves the historical v1 grid: three seed masses crossed with three $f_{\rm Edd}$ values, four constant efficiencies, and two merger boosts, for 72 reference curves in total. Seed mass uses color, $f_{\rm Edd}$ uses line style, efficiency uses line width, and merger boost uses opacity; numerical points retain the companion's original blue/orange palette. In the no-mass panel, narrow-line candidates are blue, luminous quasars are orange, and the X-ray candidate is gray. It is an assumption comparison rather than a fit. An uncertainty-filtered counterpart keeps those curves but excludes the four luminous quasars with a maximum reported black-hole-mass uncertainty above 0.7 dex. The reproducible selection is stored in the filter audit table; the canonical catalogue and rankings are unchanged.

The shared model also includes three Kerr spin cases, an illustrative photon-trapping/slim-disk efficiency above Eddington, four seed families, and a multiplicative merger boost. $B_{\rm merge} = 2$ adds $\log_{10} 2 = 0.301$ dex to the final mass; it does not double the exponential accretion rate.

4 Growth pressure: what is difficult under the baseline?

The object ranking reports required parameters instead of a single opaque compatibility score. Two central diagnostics are the required $f_{\rm Edd}$ for a $10^2\,M_\odot$ seed and the required seed mass for lifetime-average $f_{\rm Edd} = 0.3$. The global ordering is a navigation aid only. Comparisons are reported within object class and mass-comparability group, and pooled demographic inference is explicitly disallowed.

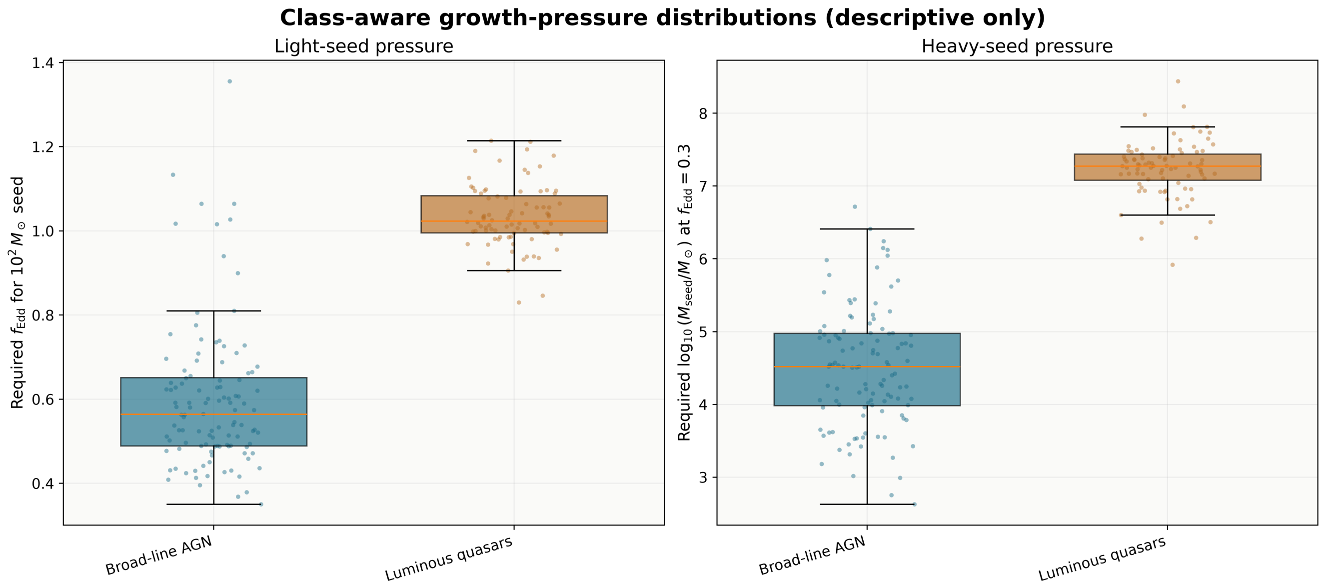

Figure 3: Class-aware distributions of the two baseline growth-pressure diagnostics. The figure is descriptive because the source selections are not a single complete population sample.

In the original v1 sample, GN-38509 requires $f_{\rm Edd} = 1.065$ for a $10^2\,M_\odot$ seed and $\log_{10}(M_{\rm seed}/M_\odot) = 6.719$ for $f_{\rm Edd} = 0.3$. In the final v3 navigation view, the luminous quasar J1148+5251 is the highest point-pressure object, with required $f_{\rm Edd} = 1.215$ and $\log_{10}(M_{\rm seed}/M_\odot) = 7.980$ under the same baseline assumptions. These values identify leverage under a coordinate system of assumptions; they do not select one formation channel.

Where to inspect it. The complete baseline tables are `results/<version>/tables/<version>_evaluation_table.csv`, `..._required_fedd_by_seed_mass.csv`, and `..._required_mseed_by_growth_assumption.csv`. Point rankings retain the source, method, evidence, eligibility, and systematic-envelope metadata beside every derived value.

5 Uncertainty: which conclusions survive reported errors?

For every eligible measurement and preferred object, the pipeline draws 10,000 deterministic samples from the reported asymmetric uncertainty in $\log_{10} M_{\rm BH}$. It recomputes the required $f_{\rm Edd}$ and seed mass for each draw, storing percentile intervals and physically meaningful threshold probabilities. Reported statistical errors are never blended with source-specific virial systematics; systematic shifts remain separately labelled envelopes.

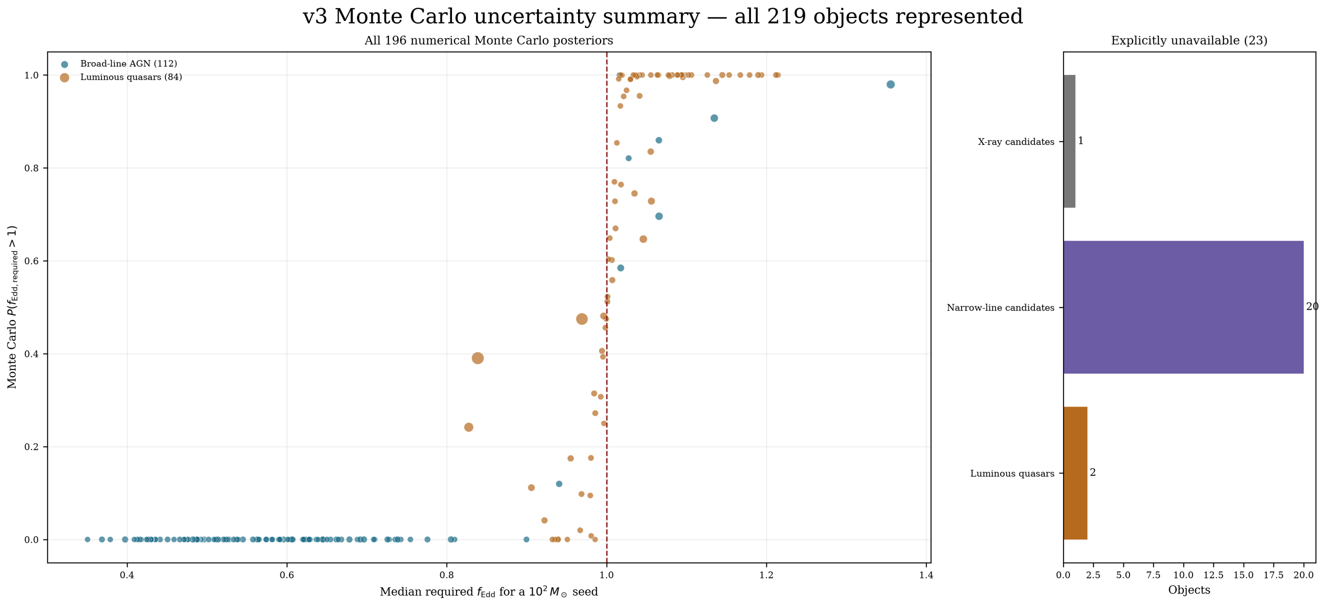

Figure 4: Monte Carlo summary for all 196 numerical v3 objects. Marker size encodes interval width; the side panel accounts for the 23 objects that do not support a numerical posterior.

The corresponding complete object atlas is `results/v3/figures/v3_all_object_monte_carlo_uncertainty.png`. In v1, GS-20057765 has a 0.980 posterior probability of requiring $f_{\rm Edd} > 1$ from a $10^2\,M_\odot$ seed under the baseline. Its tentative detection status is retained next to that numerical pressure, preventing an extreme value from being mistaken for an equally secure constraint.

The later science work also remains in every dataset version: effective two-state histories with quiescent $f_{\rm Edd} = 0$ and burst $f_{\rm Edd} = 1, 2, 3$. The stored duty cycle can exceed one, explicitly indicating that a fixed burst scenario is insufficient. Published current Eddington ratios are treated as instantaneous measurements and are not substituted for lifetime averages.

How it was obtained. `scripts/01_generate_science.ipynb` calls the shared class-aware Python engine for all three datasets. Object-stable random seeds make the Monte Carlo tables reproducible when the catalogue is expanded.

6 Compatibility: which assumption families can reach each object?

Compatibility is evaluated object by object across four seed families, three spin cases, two merger cases, and three lifetime-average Eddington ratios. Figure 5 summarizes compatible fractions within each class; the complete object-labelled matrix is stored beside it.

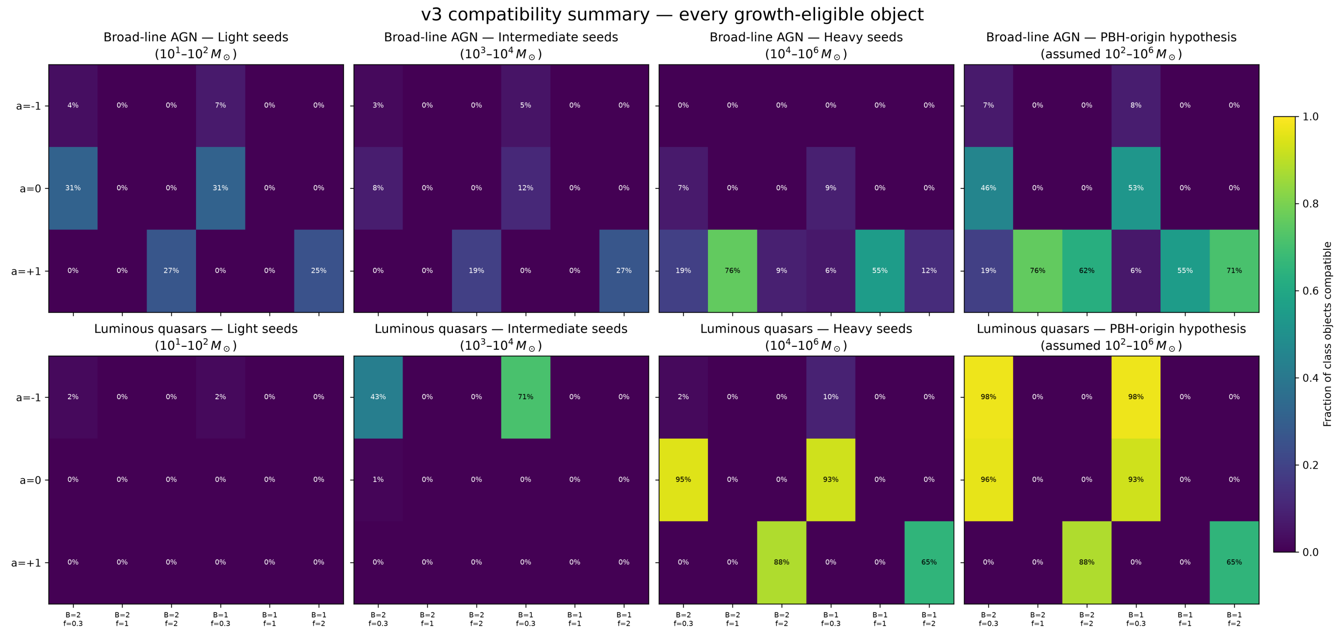

Figure 5: Class-specific compatibility fractions across seed, spin, merger, and accretion assumptions. Fractions are descriptive within the observed catalogue and are not occurrence rates.

The complete numerical table is `results/v3/tables/v3_all_object_compatibility.csv`; it contains every object-scenario combination. Unsupported objects have explicit unavailable values rather than false incompatibilities. The full visual matrix is `results/v3/figures/v3_all_object_compatibility_atlas.png`.

7 From catalogue scale to individual objects

Each catalogue object receives two canonical gallery products: a parameter-map sheet and a seed-redshift panel. The parameter sheet compares six spin/merger combinations and uses the photon-trapping efficiency prescription where appropriate. Figure 6 shows the GN-38509 example. Equivalent panels exist for every v1, v2, and v3 object.

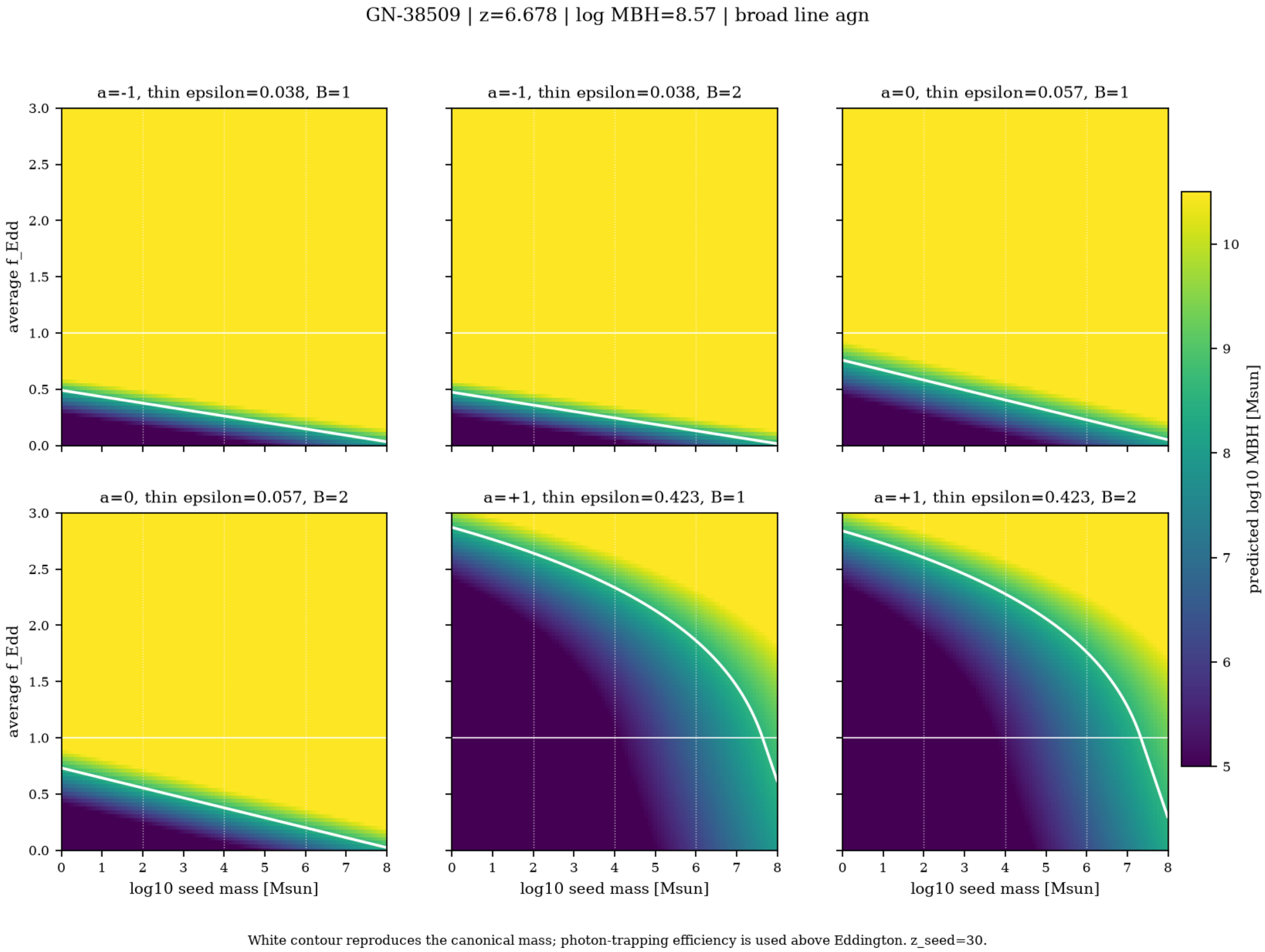

Figure 6: GN-38509 parameter-map sheet. Color gives the predicted final black-hole mass across seed mass and lifetime-average accretion rate. The observed mass contour shows which combinations reproduce the preferred measurement under each spin and merger case.

The compiled v1 map gallery demonstrates that the original 23-object dataset is complete in results rather than serving as a partial prototype. v2 and v3 use the identical panel definition at larger scale. For the 23 v3 objects without a supported mass, the same file locations contain carefully labelled status panels explaining why no numerical inference was made.

Where to browse. The per-object products are in the `fedd_mass_maps/` and `seedredshift_mass_maps/` subdirectories under `results/<version>/parameter_maps/`. Catalogue-wide galleries are in `results/<version>/figures/`. The visual coverage CSV in the version's table directory links every object to both files and distinguishes numerical products from no-inference panels.

8 Measurement choice and provenance remain visible

Thirteen eligible alternate measurements in v3 are compared against their preferred object rows. The sensitivity plot shows how a literature-measurement choice changes both $\log M_{\rm BH}$ and the derived required $f_{\rm Edd}$ without silently replacing either published value.

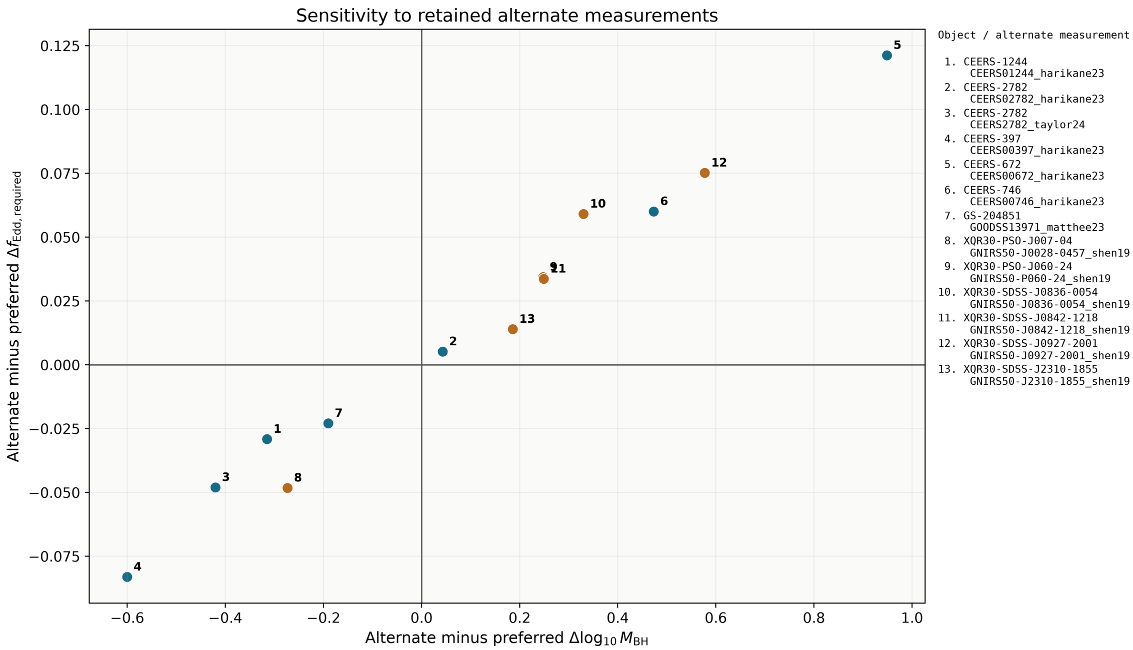

Figure 7: Sensitivity to retained alternate measurements. Numbered points map to explicit object and measurement IDs in the side key and CSV table.

Every scientific row can be traced to a source key, table, paper version, URL, DOI/archive record, extraction status, method, and caveat. Identity decisions are separate tables rather than hidden string matching. Preferred measurements control object-level results, but all measurement-level evidence states and values remain available for audit. The follow-up matrix retains all 219 objects, ranking 196 and leaving 23 explicitly unranked; the caveat summary contains one row for each of the 11 admitted source families.

9 Repository tour and exact reproduction

The shortest path through the repository is:

1. `docs/guides/versioning.md`: authoritative v1/v2/v3 membership.
2. `data/sources.md`: source-by-source extraction and caveats.
3. `data/processed/v3/v3_accreting_objects.csv`: final object view.
4. `src/datasets.py`: deterministic dataset materialization.
5. `src/models.py`: cosmology and black-hole growth equations.
6. `src/science.py`: shared rankings, uncertainty, and duty cycles.
7. `results/v3/tables/`: complete numerical paper and audit products.
8. `results/v3/figures/`: paper-ready summaries and complete atlases.
9. `results/v3/parameter_maps/fedd_mass_maps/` and `results/v3/parameter_maps/seedredshift_mass_maps/`: the full visual supplement.
10. `scripts/04_verify.ipynb`: public verification workflow.
11. `src/internal/verify_versions.py`: manifests, result inventory, exact CSV reproduction, counts, nesting, samples, compatibility, image resolution, and per-object coverage checks.

Reproduce the canonical products from the repository root:

```
mkdir -p /tmp/highz-atlas-notebooks
.venv/bin/jupyter nbconvert --to notebook --execute --output-dir=/tmp/highz-atlas-notebooks
  scripts/00_process_catalogues.ipynb
.venv/bin/jupyter nbconvert --to notebook --execute --output-dir=/tmp/highz-atlas-notebooks
  scripts/01_generate_science.ipynb
.venv/bin/jupyter nbconvert --to notebook --execute --output-dir=/tmp/highz-atlas-notebooks
  scripts/02_generate_figures.ipynb
.venv/bin/jupyter nbconvert --to notebook --execute --output-dir=/tmp/highz-atlas-notebooks
  scripts/03_generate_atlas.ipynb
.venv/bin/jupyter nbconvert --to notebook --execute --output-dir=/tmp/highz-atlas-notebooks
  scripts/04_verify.ipynb
```

The verification contract requires strict nested membership v1 $<$ v2 $<$ v3, the expected 23/112/219 object counts, 10,000 samples in Monte Carlo and duty-cycle products, complete compatibility tables, an exact result inventory, two visual panels per object, and publication-resolution PNGs. The broader regression suite and source-provenance and manuscript checks protect the ingestion details and final claims.

10 Interpretation boundary and status

The project now has a coherent paper-ready endpoint. Its central deliverable is not one favored black-hole seed model; it is a traceable map from reported measurements and evidence states to the growth assumptions each object would require. The strongest conclusions are therefore comparative:

- which objects remain demanding after reported mass uncertainty;
- which conclusions depend on source method, alternate measurement, or conditional evidence;
- which seed/accretion families are compatible under explicit spin and merger assumptions; and
- which observations would most change the physical interpretation.

The important limitations remain explicit. The catalogue mixes selection functions and cannot support pooled demographic rates. Virial systematics are not complete posterior models. Duty-cycle scenarios are effective histories, not reconstructed light curves. Candidate and disputed objects remain useful for observational triage but are not promoted to secure constraints by a high growth-pressure score.

Within those boundaries, v1 is complete in results, v2 demonstrates clean scaling across comparable objects, and v3 is the final processed catalogue and visual atlas for the source-family review cutoff of 27 August 2026. Relevant but not-yet-admitted sources are recorded in `docs/current/literature-scope.md` and require a new dataset version. Every significant paper-facing claim can be followed backward from figure to table, preferred measurement, identity decision, and source.